Exam 2 MTH 207 (Fall 2026) — J. Baggett

First Name	Last Name

Before you start. Write your name above and check that you have all 8 pages. All electronics off and out of sight — phones included, even just to tell the time, even in airplane mode. No calculators, notes, or books.

TO EARN POINTS YOU MUST SHOW YOUR WORK. Explain your reasons, justify your steps, and present your logic in an organized way. An answer with no supporting work earns no credit.

You have exactly 55 minutes. When you are told to stop you must stop **immediately** — that is in fairness to everyone. Keep your eyes on your own paper.

Geometry, if you need it. You are not expected to memorize these.

Circle	$A = \pi r^2,\ C = 2\pi r$	Sphere	$V = \frac{4}{3}\pi r^3,\ S = 4\pi r^2$
Square	$A = s^2$	Cylinder	$V = \pi r^2 h$
Rectangle	$A = \ell w$	Cone	$V = \frac{1}{3}\pi r^2 h$
Triangle	$A = \frac{1}{2}bh$	Cube	$V = s^3,\ S = 6s^2$
Right triangle	$a^2 + b^2 = c^2$		

1. [2 pts ea.] Determine whether each statement is **true** or **false**. If it is true, say why in one sentence. If it is false, explain why or give an example that disproves it.

(a) Differentiating xy with respect to x gives $x\dfrac{dy}{dx} + y$.

Solution:

TRUE. The Product Rule on x and y, with $\dfrac{d}{dx}(y) = \dfrac{dy}{dx}$ because y is a function of x. Two rules are doing work in one short expression.

(b) If f has an absolute minimum value at c, then $f'(c) = 0$.

Solution:

FALSE. c could be an endpoint, or f' could fail to exist there – $f(x) = |x|$ at 0.

2. [7 pts] Evaluate the limit with **justifications**. Simplify until you have a number.

$$\lim_{x\to 0} \frac{\sin 5x}{3x^2 - 2x}$$

Solution:

Substituting gives $\frac{0}{0}$, so the expression has to be rewritten. The only trigonometric limit we have is $\lim_{\theta\to 0} \frac{\sin\theta}{\theta} = 1$, so manufacture the matching denominator:

$$\begin{aligned}
\frac{\sin 5x}{3x^2 - 2x} &= \frac{\sin 5x}{x(3x-2)} && \text{factor } x \text{ out of the denominator} \\
&= \frac{\sin 5x}{5x} \cdot \frac{5x}{x(3x-2)} && \text{multiply and divide by } 5x \\
&= \underbrace{\frac{\sin 5x}{5x}}_{\to 1} \cdot \frac{5}{3x-2} && \text{cancel the } x
\end{aligned}$$

As $x \to 0$ the first factor tends to 1 and $3x - 2 \to -2$, so

$$\lim_{x\to 0} \frac{\sin 5x}{3x^2 - 2x} = 1 \cdot \frac{5}{-2} = \boxed{-\tfrac{5}{2}}$$

3. [9 pts] A table of values for f, g, f' and g' is given. If $h(x) = f(g(x))$, find $h'(2)$.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
1	3	4	2	5
2	1	6	3	7
3	7	8	1	9

Solution:

The Chain Rule gives $h'(x) = f'(g(x)) \cdot g'(x)$, so

$$h'(2) = f'(g(2)) \cdot g'(2)$$

Read the inner value first: the table gives $g(2) = 3$, so the outer factor is $f'(3) = 8$. The inner factor is $g'(2) = 7$.

$$h'(2) = 8 \cdot 7 = \boxed{56}$$

The usual error is using $f'(2) = 6$. The outer derivative is evaluated at the *inner function's output*, not at x.

4. [8 pts] Use a linear approximation to estimate $\sqrt[3]{8.5}$. **State the function and the point you choose**, and show your work. Leave your answer as an exact fraction.

Solution:

Choose the function and the point. The expression is a cube root, and 8 is the nearest number whose cube root is exact, so take

$$f(x) = \sqrt[3]{x} = x^{1/3}, \qquad a = 8$$

$$\begin{aligned} f(8) &= 2 \\ f'(x) &= \tfrac{1}{3}x^{-2/3} && \text{Power Rule} \\ f'(8) &= \tfrac{1}{3} \cdot \frac{1}{8^{2/3}} = \tfrac{1}{3} \cdot \tfrac{1}{4} = \tfrac{1}{12} \end{aligned}$$

The linearization at a is $L(x) = f(a) + f'(a)(x - a)$, so

$$L(x) = 2 + \tfrac{1}{12}(x - 8)$$

$$\sqrt[3]{8.5} \approx L(8.5) = 2 + \tfrac{1}{12} \cdot \tfrac{1}{2} = 2 + \tfrac{1}{24} = \mathbf{\frac{49}{24}}$$

5. [8 pts] If y depends on x according to $x^3 + y^3 = 6xy$, find $\dfrac{dy}{dx}$.

Solution:

Differentiate both sides with respect to x. Every y is a function of x, so differentiating one leaves a $\dfrac{dy}{dx}$ behind.

$$\begin{aligned} \frac{d}{dx}\left(x^3 + y^3\right) &= \frac{d}{dx}\left(6xy\right) \\ 3x^2 + 3y^2\frac{dy}{dx} &= 6y + 6x\frac{dy}{dx} && \text{Chain Rule on } y^3;\ \text{Product Rule on } 6xy \end{aligned}$$

Collect the $\dfrac{dy}{dx}$ terms on one side and everything else on the other:

$$\begin{aligned} 3y^2\frac{dy}{dx} - 6x\frac{dy}{dx} &= 6y - 3x^2 \\ \left(3y^2 - 6x\right)\frac{dy}{dx} &= 6y - 3x^2 && \text{factor out } \tfrac{dy}{dx} \\ \frac{dy}{dx} &= \frac{6y - 3x^2}{3y^2 - 6x} = \mathbf{\frac{2y - x^2}{y^2 - 2x}} \end{aligned}$$

6. Find the derivative of each function below.

(a) *(5 pts)* $f(x) = \dfrac{2x^2 - 1}{x^3 + 4}$

Solution:

Quotient Rule, $\left(\dfrac{u}{v}\right)' = \dfrac{u'v - uv'}{v^2}$, with $u = 2x^2 - 1$ and $v = x^3 + 4$:

$$\begin{aligned} f'(x) &= \frac{(4x)\left(x^3+4\right) - \left(2x^2-1\right)\left(3x^2\right)}{\left(x^3+4\right)^2} \\ &= \frac{4x^4 + 16x - \left(6x^4 - 3x^2\right)}{\left(x^3+4\right)^2} \\ &= \boxed{\frac{\mathbf{-2x^4 + 3x^2 + 16x}}{\mathbf{\left(x^3+4\right)^2}}} \end{aligned}$$

(b) *(5 pts)* $y = 4^x \log_3(x)$

Solution:

A product, so the Product Rule runs first. The two pieces need the base-a rules:

$$\frac{d}{dx}a^x = a^x \ln a, \qquad \frac{d}{dx}\log_a x = \frac{1}{x \ln a}$$

$$\begin{aligned} y' &= (4^x \ln 4)\log_3(x) + 4^x \cdot \frac{1}{x \ln 3} \\ &= \boxed{\mathbf{4^x \ln 4}\ \log_{\mathbf{3}}(\mathbf{x}) + \frac{\mathbf{4^x}}{\mathbf{x \ln 3}}} \end{aligned}$$

The $\ln 4$ and the $\ln 3$ are part of the answer, not optional decoration — and note the ln is upstairs for a^x and downstairs for $\log_a x$.

(c) *(5 pts)* $y = x^{\cos x}$

Solution:

Neither the base nor the exponent is constant, so no rule applies directly. Take logarithms of both sides first.

$$\begin{aligned} \ln y &= \cos x \ln x \\ \frac{1}{y}\cdot\frac{dy}{dx} &= -\sin x \ln x + \cos x \cdot \frac{1}{x} \qquad \text{Chain Rule on the left, Product Rule on the right} \\ \frac{dy}{dx} &= y\left(\frac{\cos x}{x} - \sin x \ln x\right) \\ &= \boxed{\mathbf{x}^{\cos \mathbf{x}}\left(\frac{\cos \mathbf{x}}{\mathbf{x}} - \sin \mathbf{x} \ln \mathbf{x}\right)} \end{aligned}$$

7. Derive the following derivative rules.

(a) *(7 pts)* Derive the derivative rule for $y = \sec x$. (You may use the derivative rules for sine and cosine.)

Solution:

Write the secant as a quotient of things we can differentiate, then use the Quotient Rule.

$$\begin{aligned} y = \sec x &= \frac{1}{\cos x} \\ y' &= \frac{(0)(\cos x) - (1)(-\sin x)}{\cos^2 x} && \text{Quotient Rule} \\ &= \frac{\sin x}{\cos^2 x} \\ &= \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x} && \text{split the fraction} \\ &= \sec \mathbf{x} \tan \mathbf{x} \end{aligned}$$

(b) *(7 pts)* Derive the derivative rule for $y = \tan^{-1} x$. (You may use the derivative rules for the trig functions, but **not** any derivative rules for inverse trig functions.)

Solution:

Let $y = \tan^{-1} x$. The point of the substitution is to get rid of the inverse: taking the tangent of both sides turns it into something we can differentiate.

$$\begin{aligned} \tan y &= x \\ \sec^2 y \cdot \frac{dy}{dx} &= 1 && \text{differentiate both sides; Chain Rule on the left} \\ \frac{dy}{dx} &= \frac{1}{\sec^2 y} \end{aligned}$$

The answer still mentions y, so convert back using the Pythagorean identity $\sec^2 y = 1 + \tan^2 y$ together with $\tan y = x$:

$$\frac{dy}{dx} = \frac{1}{1 + \tan^2 y} = \frac{\mathbf{1}}{\mathbf{1 + x^2}}$$

8. [15 pts] Water is poured into a cone-shaped tank at a rate of 2 cubic feet per minute. The cone points downward; it is 10 feet tall and its circular top has radius 5 feet. How fast is the depth of the water rising when the water is 4 feet deep? **Include units**, and leave your answer exact — a π in it is fine.

Solution:

Let h be the depth of the water and r the radius of its circular surface, both functions of time t. We are given $\dfrac{dV}{dt} = 2$ and want $\dfrac{dh}{dt}$ at the instant $h = 4$.

The water itself forms a cone, so $V = \dfrac{1}{3}\pi r^2 h$. That has two changing lengths in it, so one has to go. The water cone and the tank are similar, and the tank is 10 ft tall with radius 5 ft, so

$$\frac{r}{h} = \frac{5}{10} = \frac{1}{2} \qquad \Longrightarrow \qquad r = \frac{h}{2}$$

This is a relation that holds at *every* instant, not a value at one moment, so it may be substituted now — and that is what makes the problem one-variable:

$$V = \frac{1}{3}\pi \left(\frac{h}{2}\right)^2 h = \frac{\pi h^3}{12}$$

Now differentiate with respect to t:

$$\frac{dV}{dt} = \frac{\pi}{12} \cdot 3h^2 \cdot \frac{dh}{dt} = \frac{\pi h^2}{4} \cdot \frac{dh}{dt}$$

Only now put the numbers in, with $\dfrac{dV}{dt} = 2$ and $h = 4$:

$$2 = \frac{\pi (4)^2}{4} \cdot \frac{dh}{dt} = 4\pi \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{\mathbf{1}}{\mathbf{2}\pi} \text{ ft/min}$$

Substituting $h = 4$ before differentiating is the common fatal error: it freezes the depth, and its derivative becomes 0.

9. [12 pts] Let $f(x) = 2x^3 - 3x^2 - 12x + 1$. Find the absolute maximum and absolute minimum values of f on the interval $[-2, 3]$, and say where each occurs.

Solution:

f is a polynomial, so it is continuous on the closed interval $[-2, 3]$ and the Extreme Value Theorem guarantees that both an absolute maximum and an absolute minimum exist. By the Closed Interval Method each one occurs at a critical number or at an endpoint.

1. Find the critical numbers in $[-2, 3]$.

$$\begin{aligned} f'(x) &= 6x^2 - 6x - 12 \\ &= 6\left(x^2 - x - 2\right) \\ &= 6(x-2)(x+1) \end{aligned}$$

f' is defined everywhere, so the critical numbers are where $f' = 0$: $x = -1$ and $x = 2$. Both lie in $[-2, 3]$.

2. Make the table, with the endpoints in it as well.

x	$f(x)$	why it is on the list
-2	-3	left endpoint
-1	8	critical number
2	-19	critical number
3	-8	right endpoint

3. By the Extreme Value Theorem the absolute maximum and minimum must be in that table, so compare the four values.

4. f has an absolute **maximum value of 8 at $x = -1$**, and an absolute **minimum value of -19 at $x = 2$**.

10. [8 pts] Verify that $f(x) = x^3 - x$ on $[0, 2]$ satisfies the hypotheses of the Mean Value Theorem (clearly state why each hypothesis is satisfied) and find all numbers c that satisfy the conclusion of the Mean Value Theorem. Give **exact** values.

Solution:
The hypotheses. $f(x) = x^3 - x$ is a polynomial.

- Polynomials are continuous everywhere, so f is continuous on the *closed* interval $[0, 2]$.
- Polynomials are differentiable everywhere, so f is differentiable on the *open* interval $(0, 2)$.

Both hypotheses hold, so the Mean Value Theorem applies.
The conclusion. It promises at least one c in $(0, 2)$ with

$$f'(c) = \frac{f(2) - f(0)}{2 - 0}$$

The right-hand side is the slope of the secant line:

$$f(2) = 8 - 2 = 6, \qquad f(0) = 0, \qquad \frac{f(2) - f(0)}{2 - 0} = \frac{6 - 0}{2} = 3$$

Set $f'(c)$ equal to it and solve:

$$\begin{aligned} f'(x) &= 3x^2 - 1 \\ 3c^2 - 1 &= 3 \\ 3c^2 &= 4 \\ c^2 &= \frac{4}{3} \\ c &= \pm\frac{2}{\sqrt{3}} \end{aligned}$$

The MVT only promises a c *inside* $(0, 2)$, and $-\dfrac{2}{\sqrt{3}}$ is not in that interval. So the only such number is

$$c = \frac{2}{\sqrt{3}} = \boxed{\frac{2\sqrt{3}}{3}} \approx 1.15$$

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